V11.1B V2 — Planck-Area Writes, Lag Stress, and Fluid Coefficient Map

From Boundary Accounting to Continuum Fluid Coefficients

Status	Bridge derivation note with computational support and coefficient map
Parent note	V11.1 — The Planck-Area Accounting Lemma
Companion result	V11.7 Fluid Sector
Support toys	1D scalar, 2D scalar, 2D tensor, V11.1D multipole diagnostic

Core bridge:

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Planck-area writes
→ finite settlement lag
→ scalar/tensor lag stress
→ V11.7 fluid correction
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This PDF is a standalone V11.1B bridge note. It keeps the original V11.1 Planck-area lemma as the parent foundation and adds the missing finite-settlement layer required to connect that foundation to the V11.7 fluid-stress result.

0. Abstract

V11.1 established the Planck-area accounting lemma: one complete maintenance tick demands one elementary boundary capacity increment. In its compact form, the single-particle write-demand rate is:

$$\dot{A}_{tick} = \xi\, \ell_P^2\, (m c^2 / h) = \xi\, Gm / (2\pi c).$$

V11.1B develops the missing middle layer between that discrete accounting statement and the V11.7 fluid sector. The new object is not the instantaneous write demand alone, but the difference between demanded and settled boundary accounting. That difference is the lag residual:

$$\Lambda(x,t) = A_{demand}(x,t) - A_{settled}(x,t).$$

When the write pattern changes faster than the boundary/geometry can settle, Λ becomes nonzero. Gradients and anisotropies of Λ coarse-grain into effective scalar and tensor lag stresses. The purpose of this note is to define the bridge variables, derive the natural candidate terms, and record the small numerical support tests showing that scalar and tensor lag behavior survive 1D and 2D promotion.

1. What V11.1 Already Establishes

V11.1 establishes that the Planck area is a boundary capacity unit, not a bulk lattice cell. The ledger is discrete; curvature is the smooth coarse-grained response to many boundary accounting increments.

$$\text{one tick} \Rightarrow \delta A = \xi\, \ell_P^2$$

$$\text{many local ticks} \Rightarrow \nabla^2\psi = (4\pi G h/c^4)\, \rho_{ops} \Rightarrow \text{gravity}$$

$$\text{all ticks globally} \Rightarrow N_H = I_H \Rightarrow H_{SFT}^2/H_{std}^2 = 1/(4\pi^2)$$

The next question is what happens when local write demand changes faster than settlement can complete. That question introduces lag stress.

2. From Point Demand to Local Demand Density

For a single rest particle, the Planck-area accounting lemma gives:

$$\dot{A}_{tick} = \xi\, Gm / (2\pi c).$$

For a small volume element with mass density ρ_m, $dm = \rho_m(x,t)dV$, so the local area-demand rate becomes:

$$d\dot{A}_{demand} = (\xi G / 2\pi c)\, \rho_m(x,t)\, dV.$$

For a general relativistic fluid, the operation density used by V11.1 is:

$$\rho_{ops} = (\varepsilon + 3p) / h.$$

The write-demand density is then:

$$\dot{a}_{demand}(x,t) = \xi\ \ell_P^2\ \rho_{ops}(x,t).$$

For pressureless matter this reduces to:

$$\dot{a}_{demand} = \xi\ \ell_P^2\ (\rho_m c^2/h) = \xi G \rho_m/(2\pi c).$$

Thus the point-particle lemma becomes a local density statement:

$$\dot{a}_{demand}(x,t) \propto \rho_{ops}(x,t).$$

3. Demand Is Not Settlement

V11.1 describes how ticks demand boundary capacity. But a demand is not automatically a settled geometric response. Define the integrated demand:

$$A_{demand}(x,t) = \int^t \dot{a}_{demand}(x,t')\, dt'.$$

Define A_settled as the part of the demand already absorbed into the coarse-grained geometry. The lag residual is:

$$\Lambda(x,t) = A_{demand}(x,t) - A_{settled}(x,t).$$

If settlement is instantaneous, $\Lambda = 0$. If demand changes faster than settlement, $\Lambda \neq 0$. This is the central new object of V11.1B.

4. Relaxation Law for Settlement

The simplest causal settlement law is first-order relaxation:

$$\partial_t A_{settled} = (A_{demand} - A_{settled}) / \tau_{lag} = \Lambda / \tau_{lag}.$$

Differentiating the residual gives the minimal scalar lag equation:

$$\partial_t \Lambda = \dot{a}_{demand} - \Lambda/\tau_{lag}.$$

A purely local relaxation law is incomplete because settlement cannot propagate faster than c. A minimal causal-diffusive form is:

$$\partial_t \Lambda + \Lambda/\tau_{lag} - D_\Lambda \nabla^2\Lambda = \dot{a}_{demand}.$$

A more explicitly causal telegraph-like form would be:

$$\tau_{lag}\, \partial_t^2\Lambda + \partial_t\Lambda - c_{lag}^2\nabla^2\Lambda + \Lambda/\tau_{lag} = \dot{a}_{demand},$$
$$\text{with } c_{lag} \le c.$$

The minimal program does not require fixing the final propagation law immediately. It requires acknowledging that settlement is not instantaneous, so unmet demand can have gradients, memory, and anisotropy.

5. From Lag Residual to Effective Force

The scalar lag residual can influence local motion through its gradient. The simplest scalar lag acceleration is:

$$a_\Lambda = -\kappa_1 \nabla\Lambda.$$

Uniform lag residual produces no local force. Gradients in lag residual produce a force-like smoothing term tied to unsettled accounting imbalance:

$$f_{scalar} \sim -\nabla\Lambda.$$

This is not ordinary gravity itself. It is the residual between demanded and settled accounting. It only exists when the write pattern is not fully settled.

6. Tensor Lag: Directional Settlement Mismatch

V11.7 found that scalar and tensor contributions separate meaningfully. Therefore V11.1B must define a directional lag object. In two dimensions, the natural trace-free Hessian is:

$$Q_{ij} = \partial_i\partial_j\Lambda - (1/2)\delta_{ij} \nabla^2\Lambda.$$

In three dimensions the corresponding trace-free object is:

$$\Lambda_{ij}^{TF} = (\partial_i\partial_j - (1/3)\delta_{ij}\nabla^2)\Lambda.$$

A candidate lag stress is:

$$\sigma_{ij}^{lag} = -\eta_\Lambda \Lambda_{ij}^{TF}.$$

The corresponding force density is:

$$f_i^{tensor} = \partial_j \sigma_{ij}^{lag}.$$

This gives the bridge:

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anisotropic settlement mismatch
→ trace-free lag stress
→ shear-like correction
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7. Relation to Fluid Variables

In a continuum fluid, the demand field follows the operation density. For a fluid with velocity u, operation density is transported:

$$\partial_t \rho_{ops} + \nabla\cdot(\rho_{ops} u) = \text{sources/sinks}.$$

A moving density field creates time-dependent demand gradients, so the lag residual responds to:

$$\partial_t \Lambda = S[\rho_{ops},u] - \Lambda/\tau_{lag}.$$

The lowest-order continuum corrections are built from $\nabla\Lambda$, $\nabla^2\Lambda$, Λ_{ij}^{TF}, and their divergence. These are exactly the ingredients needed to connect V11.1 to the V11.7 scalar/tensor fluid split.

8. Mapping to the V11.7 Fluid Sector

V11.7 found that lag-like corrections can alter fluid behavior, that scalar and tensor contributions separate meaningfully, and that the effect is not reducible to ordinary numerical viscosity. V11.1B supplies an origin for those terms.

V11.7 object	V11.1B interpretation
scalar smoothing	gradient/curvature of scalar lag residual Λ
shear-memory damping	trace-free directional settlement mismatch Λ_{ij}^{TF}
compressive smoothing	isotropic curvature component $\nabla^2\Lambda$
viscosity independence	lag stress is settlement memory, not grid viscosity
control parameter	strength and response time of settlement residual

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V11.7 lag stress = continuum response of delayed Planck-area settlement.
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9. Why This Is Not Ordinary Viscosity

Ordinary viscosity dissipates velocity gradients:

$$\sigma_{ij}^{visc} \sim \eta(\partial_i u_j + \partial_j u_i - (2/3)\delta_{ij} \nabla\cdot u).$$

Lag stress instead responds to unsettled accounting gradients:

$$\sigma_{ij}^{lag} \sim -\eta_\Lambda(\partial_i\partial_j - (1/3)\delta_{ij}\nabla^2)\Lambda.$$

The difference is operational: viscosity is a local transport coefficient of the material medium, while lag stress is geometric/accounting memory of delayed settlement. The two may look similar in some continuum limits, but they do not have the same physical source.

10. Computational Support Layer

After the bridge equations were drafted, four small numerical diagnostics were run. The purpose was not to reproduce the full V11.7 fluid suite, but to check the sign, shape, memory, and tensor behavior of the proposed lag residual.

10.1 1D scalar sign/shape toy

A moving 1D operation-demand pulse was evolved with:

$$\partial_t\Lambda = S - \Lambda/\tau_\Lambda + D_\Lambda \, \partial_x^2\Lambda,$$
$$f_\Lambda = -\partial_x\Lambda.$$

The 1D test passed three gates: Λ trails the moving source for finite τ_Λ, the force proxy collapses as $\tau_\Lambda \to 0$, and the lag force is shifted relative to the instantaneous source-gradient force.

10.2 2D scalar lag toy

The same equation was promoted to two dimensions and tested on a moving blob, a rotated moving blob, and crossing blobs. The residual trailed the moving source along the velocity vector, vanished in the fast-settlement limit, rotated correctly with the source, and stored history in crossing geometries.

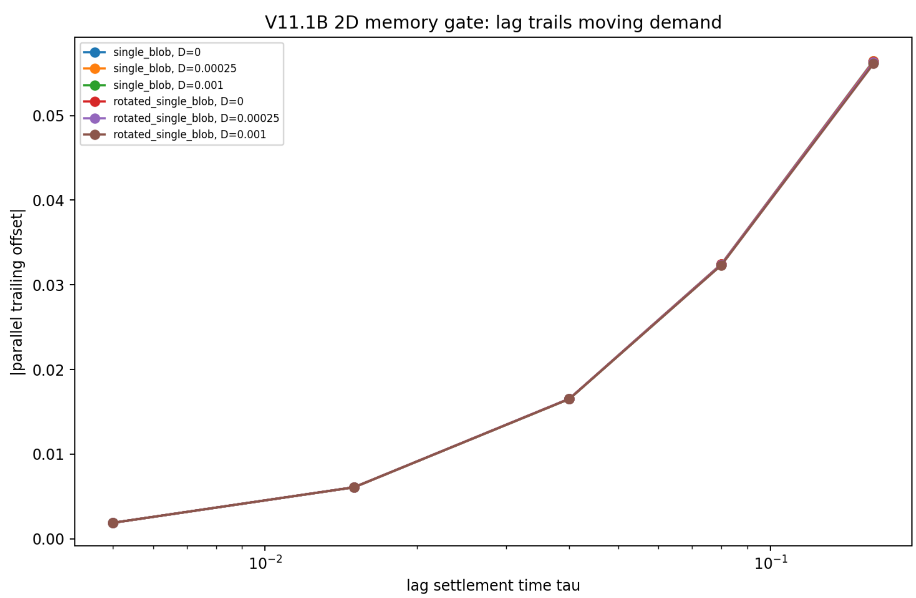

Figure 1. The 2D scalar memory gate. The lag residual trails a moving demand source more strongly as settlement time increases.

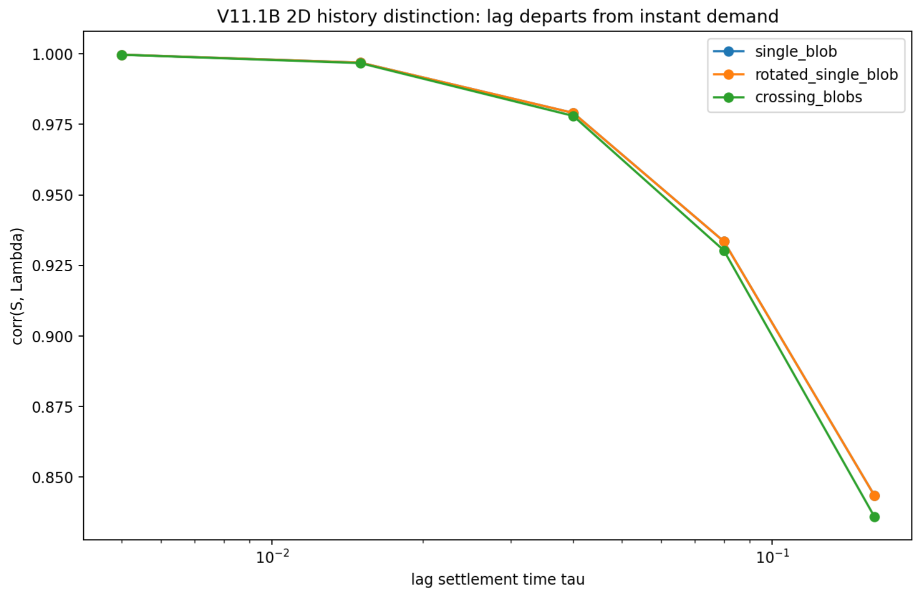

Figure 2. The lag residual departs from the instantaneous demand map as τ_Λ grows, indicating memory rather than instantaneous smoothing.

10.3 2D tensor lag toy

The tensor diagnostic computed the trace-free Hessian of Λ and its divergence:

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Q_ij = ∂_i∂_jΛ - (1/2)δ_ij∇²Λ,
f_i^tensor = ∂_j Q_ij.
```

The main gate asked whether anisotropic residuals activate the tensor channel more strongly than round residuals. They did.

Shape	Tensor force vs round	Q_rms vs round	Anisotropy vs round	Gate
elongated blob	2.27x	1.74x	2.65x	pass
rotated elongated blob	2.27x	1.74x	2.65x	pass
crossing blobs	1.42x	1.42x	2.85x	pass

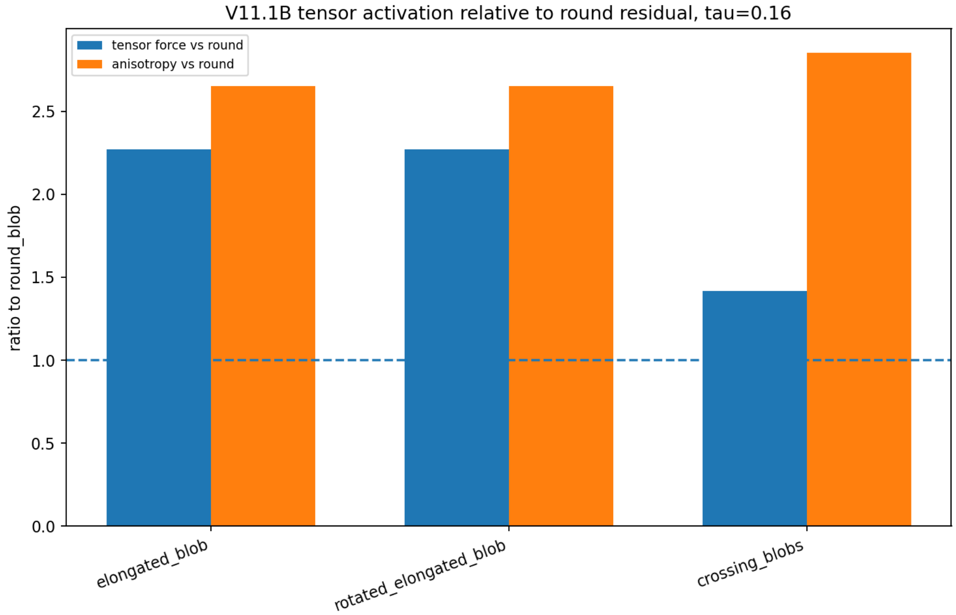

Figure 3. Raw tensor activation relative to round residuals at τ_Λ = 0.16. Elongated and crossing residuals activate the tensor channel above the round baseline.

10.4 V11.1C caution: radial shear-excess was not clean

A follow-up diagnostic attempted to subtract a radialized isotropic baseline. This was informative but not clean. A moving elongated residual contains monopole, dipole, and quadrupole structure. Radializing about the centroid mixes those modes and can over-subtract the very shear signal being measured.

The conservative conclusion is:

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raw tensor activation passes; radial shear-excess isolation remains open.
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10.5 V11.1D closure: angular multipoles

V11.1D replaced radial subtraction with angular multipole decomposition. Around the lag residual centroid:

$$\Lambda(r,\theta) = a_0(r) + \Sigma_m \, [a_m(r)\cos(m\theta) + b_m(r)\sin(m\theta)]$$

with mode amplitude:

$$A_m(r) = \mathrm{sqrt}(a_m(r)^2 + b_m(r)^2).$$

The physical content of the first modes is:

- $m = 0$: monopole, the isotropic radial profile.
- $m = 1$: dipole, the trailing memory of the moving source.
- $m = 2$: quadrupole, the shear-like anisotropic lag channel.
- $m \geq 3$: higher angular structure from complex histories.

Shape	Tensor/scalar vs round	P_2 fraction	P_2 vs round	$Phase_2$	Gate
round	1.000	0.0509	1.000	44.96°	control
elongated	1.643	0.2798	5.49	45.00°	pass
rotated elongated	1.644	0.2797	5.49	120.01°	pass
crossing	1.006	0.5428	10.66	0.00°	P_2 pass

V11.1D produced the clean result V11.1C was trying to isolate: quadrupole power tracks shear geometry, and the m=2 phase rotates with source orientation.

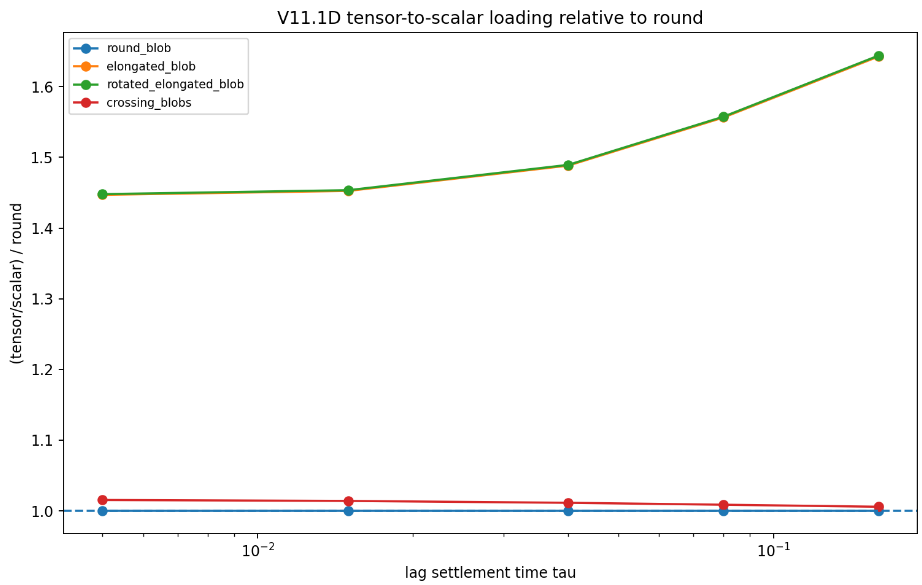

Figure 4. Tensor-to-scalar loading relative to the round baseline. Elongated and rotated elongated residuals preferentially load the tensor channel.

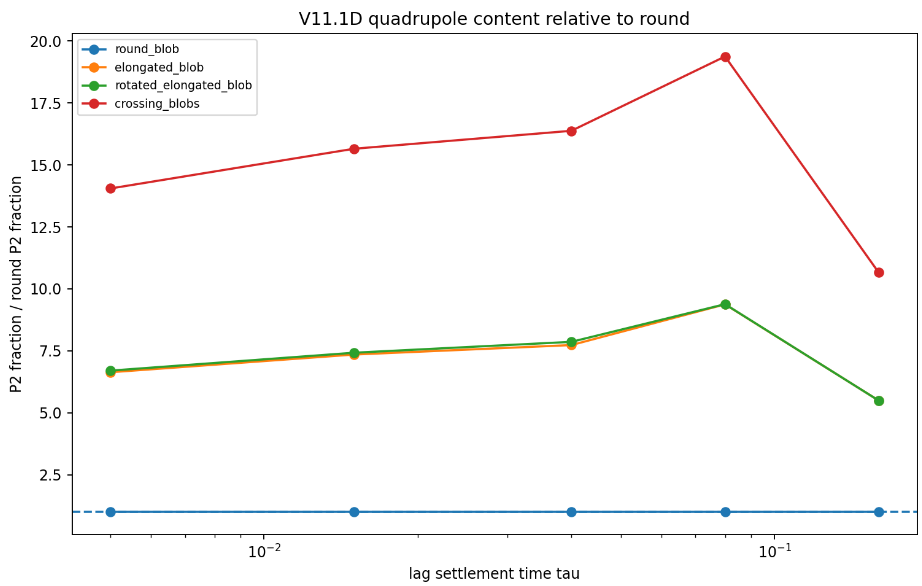

Figure 5. Quadrupole content relative to the round baseline. The m=2 mode is strongly enhanced for elongated and crossing residual histories.

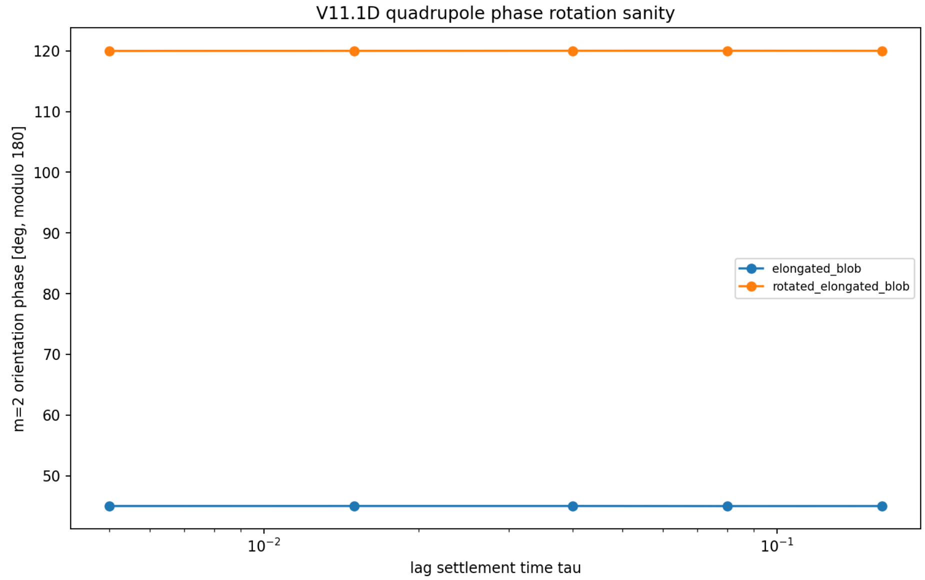

Figure 6. Quadrupole phase rotation sanity. The elongated residual sits at 45° while the rotated residual sits at 120°, matching the 75° source-orientation shift.

11. Sign Expectations and Falsification Gates

The bridge has several sign and behavior expectations:

- Over-demanded regions accumulate positive lag residual.
- Gradients in lag residual oppose unresolved compression or runaway concentration.
- Anisotropic lag residual activates a trace-free tensor/shear response.
- Uniform lag residual produces no local force.
- The correction vanishes in the instantaneous-settlement limit.

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τ_lag → 0 ⇒ Λ → 0 ⇒ lag correction → 0.
```

If a candidate fluid correction does not vanish as settlement becomes instantaneous, it is not the V11.1B lag-stress bridge.

12. What This Note Establishes

- V11.1 supplies the discrete Planck-area write demand.
- Local operation density defines a write-demand density.
- Demand and settlement differ at finite response time.
- The difference is the lag residual Λ.
- Scalar gradients of Λ produce scalar lag forces.
- Trace-free derivatives of Λ produce tensor/shear lag response.
- 1D and 2D toys confirm memory, vanishing-lag behavior, rotational sanity, and crossing-history behavior.

• V11.1D confirms that m=2 quadrupole power tracks shear geometry and rotates with source orientation.

The key scalar bridge is:

$$\partial_t\Lambda = \dot{a}_{demand} - \Lambda/\tau_{lag} + D_\Lambda\nabla^2\Lambda.$$

The key tensor bridge is:

$$\sigma_{ij}^{lag} \sim -\eta_\Lambda(\partial_i\partial_j - (1/3)\delta_{ij}\nabla^2)\Lambda.$$

13. What This Note Does Not Establish

• The exact value of τ_{lag}.

• The exact value of D_Λ.

• The exact normalization of α_Λ or η_Λ.

• A full derivation of V11.7 coefficients from first principles.

• A proof that the V11.7 fluid correction is unique.

• A covariant relativistic lag-stress tensor.

• A full numerical reproduction of the V11.7 robustness suite.

Those are the next gates. V11.1B establishes the bridge structure and the first scalar/tensor toy support layer.

14. Coefficient Map to the V11.7 Fluid Sector

V11.1E mapped the V11.1B lag-settlement variables onto the V11.7 scalar/tensor fluid coefficients. This closes the bridge from discrete boundary accounting to the computational fluid-stress sector.

The parent V11.1B lag equation is:

$$\partial_t\Lambda = \beta_\Lambda\, \rho_{ops} - \Lambda/\tau_\Lambda + D_\Lambda\, \nabla^2\Lambda.$$

14.1 Scalar coefficient dictionary

The V11.7 scalar channel uses a scalar lag-pressure field χ:

$$\partial_t\chi = D_\chi\nabla^2\chi + \alpha_\chi(\rho - \langle\rho\rangle) - \chi/\tau_\chi.$$

with scalar force:

$$f_\chi = -\varepsilon_\chi\, \nabla\chi.$$

The scalar dictionary is:

V11.1B object	V11.7 object	Interpretation
Λ	χ	scalar continuum representative of lag residual

V11.1B object	V11.7 object	Interpretation
β_Λ	α_χ	source loading from operation/density demand
τ_Λ	τ_χ	settlement memory time
D_Λ	D_χ	residual spreading / nonlocality
scalar force coupling	ε_χ	coupling in $-\varepsilon_\chi \nabla\chi$

The effective scalar control parameter is:

$$\Lambda_\chi = \varepsilon_\chi \alpha_\chi \tau_\chi / D_\chi.$$

14.2 Tensor coefficient dictionary

The V11.7 tensor channel uses a trace-free shear-memory field Q:

$$\partial_t Q = D_Q \nabla^2 Q + \alpha_Q S - Q/\tau_Q.$$

with tensor force:

$$f_Q = \varepsilon_Q \nabla \cdot Q.$$

The tensor dictionary is:

V11.1B object	V11.7 object	Interpretation
trace-free lag memory	Q	continuum tensor/shear memory field
τ_Λ	τ_Q	tensor settlement memory time
D_Λ	D_Q	tensor residual spreading
tensor source loading	α_Q	loading from trace-free strain source S
tensor force coupling	ε_Q	coupling in $\varepsilon_Q \nabla\cdot Q$

The effective tensor control parameter is:

$$\Lambda_Q = \varepsilon_Q \alpha_Q \tau_Q / D_Q.$$

14.3 Identity checks

V11.1E verified the control-parameter identities numerically to roundoff precision:

Sector	Identity	Max absolute error
1D scalar	$\Lambda = \varepsilon \alpha \tau_\chi / D_\chi$	4.44×10^{-16}
2D scalar χ	$\Lambda_\chi = \varepsilon_\chi \alpha_\chi \tau_\chi / D_\chi$	4.72×10^{-16}
2D tensor Q	$\Lambda_Q = \varepsilon_Q \alpha_Q \tau_Q / D_Q$	4.44×10^{-16}

This means the V11.7 control parameters are not arbitrary knobs. They are effective continuum packages:

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effective strength ~ (force coupling × source loading × memory time) / spreading.
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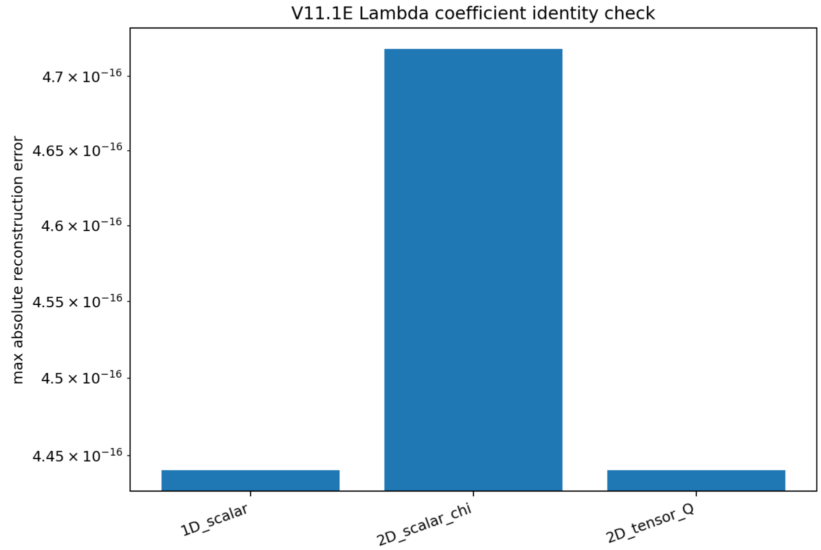

Figure 7. Lambda coefficient identities reconstruct to numerical roundoff.

14.4 Scalar response

The V11.1E scalar-only aggregate showed that increasing Λ_χ primarily changes density/compression structure and barely touches vorticity directly.

Metric slope vs Λ_χ	Value
density ratio	-3.72 × 10^-2
divergence ratio	+3.07 × 10^-2
vorticity ratio	+1.03 × 10^-3

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scalar lag → compressive/density redistribution.
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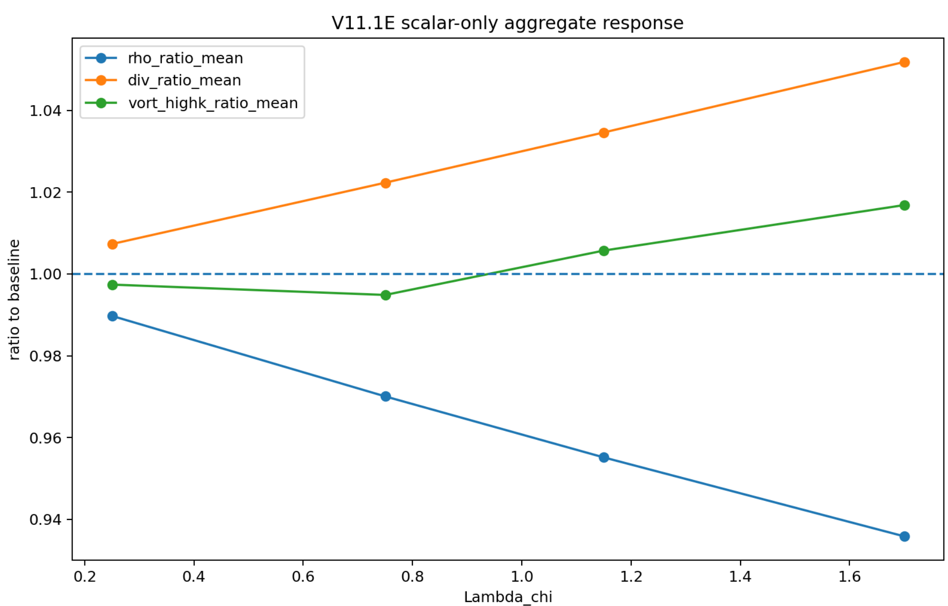

Figure 8. Scalar-only aggregate response. The scalar channel mainly redistributes density/compressive structure.

14.5 Tensor response

The V11.1E tensor-only aggregate showed that increasing Λ_Q damps shear, vorticity, enstrophy, and strain structure.

Metric slope vs Λ_Q	Value
vorticity ratio	-9.34×10^{-2}
enstrophy ratio	-1.63×10^{-1}
strain ratio	-9.11×10^{-2}

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tensor lag → trace-free shear-memory damping.
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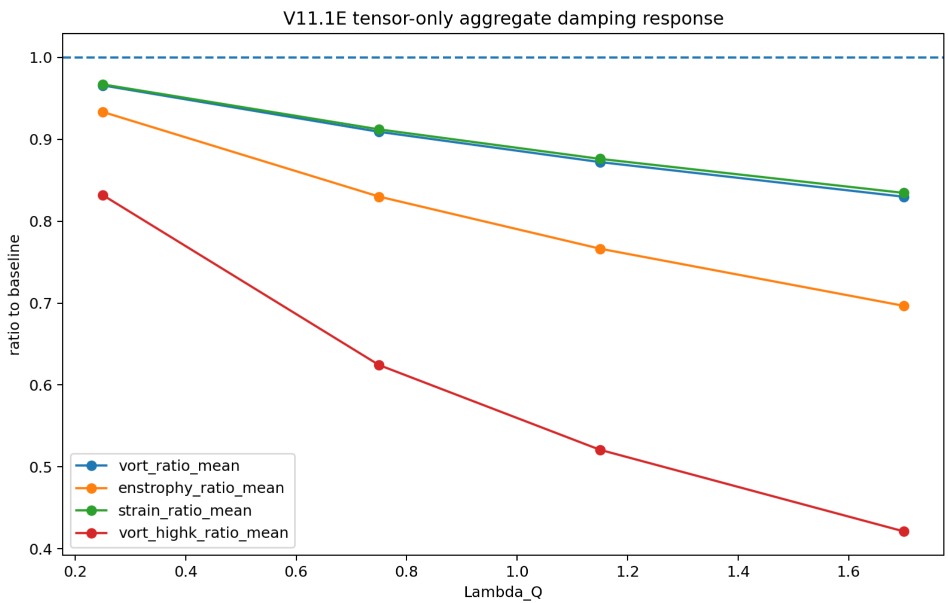

Figure 9. Tensor-only aggregate damping response. The tensor channel suppresses vorticity, enstrophy, strain, and high-k structure.

14.6 High-k preference

For a Fourier mode, the lag response has the relaxation-diffusion form:

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Λ_k = β_Λ ρ_k / (τ_Λ^-1 + D_Λ k² - iω).
```

The scalar force scales as:

```
|f_scalar| ~ k |Λ_k|.
```

The tensor force, as a divergence of a trace-free Hessian, scales as:

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|f_tensor| ~ k³ |Λ_k|.
```

Therefore:

```
|f_tensor| / |f_scalar| ~ k².
```

This predicts a tensor preference for small-scale/high-k structure. V11.1E confirmed it: at Λ_Q = 1.70, the total vorticity ratio was 0.830, while the high-k vorticity ratio was 0.421.

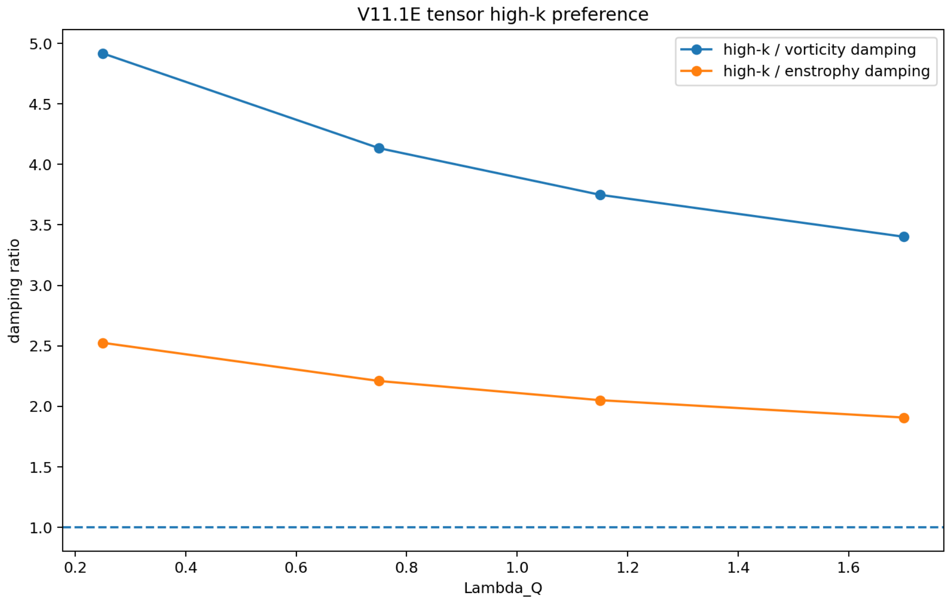

Figure 10. Tensor high-k preference. High-k vorticity is suppressed much more strongly than total vorticity.

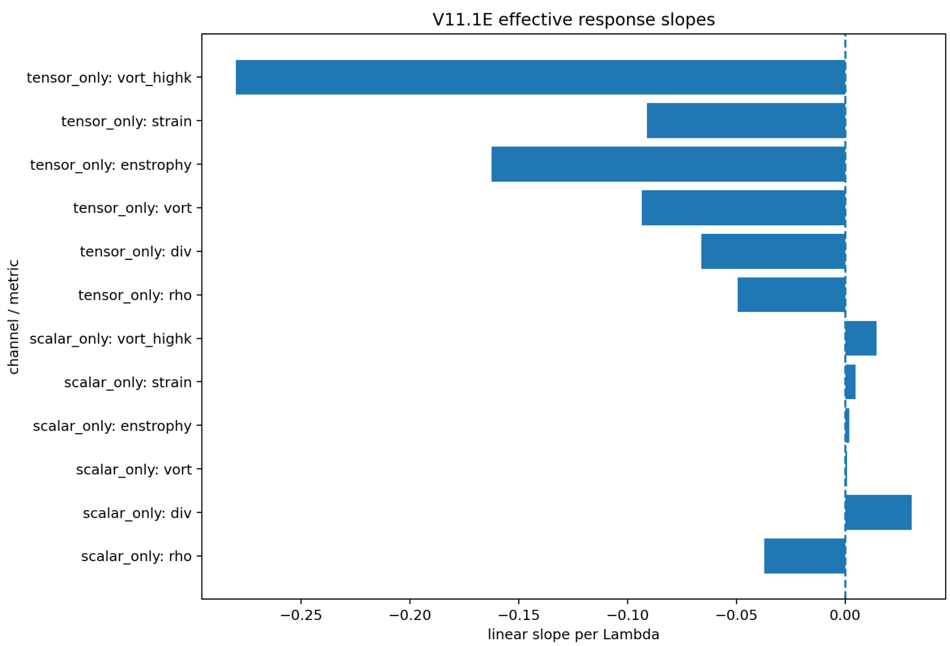

Figure 11. Effective response slopes. Scalar and tensor channels act on different fluid observables.

14.7 Combined scalar/tensor response

V11.1E also tested whether the combined V11.7 scalar+tensor channel behaves approximately like independent scalar and tensor responses:

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R_combined ≈ R_scalar R_tensor.
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Across the tested metrics, the mean absolute multiplicative residual was 6.13 × 10^-3. This supports the V11.7 scalar/tensor separation:

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scalar and tensor lag channels are mostly separable continuum responses.
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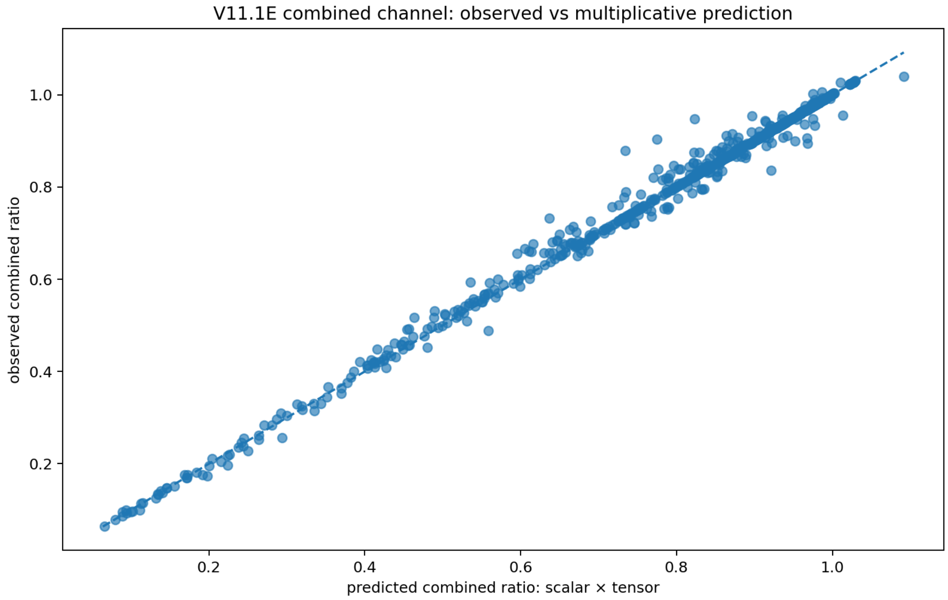

Figure 12. Combined-channel observed response versus scalar × tensor prediction. Residuals are small.

15. What This Note Establishes

• V11.1 supplies the discrete Planck-area write demand.

• V11.1B supplies the finite-settlement residual Λ.

• Scalar gradients of Λ produce scalar lag forces.

• Trace-free derivatives of Λ produce tensor/shear lag response.

• 1D and 2D scalar toys confirm memory, trailing, and vanishing-lag behavior.

• 2D tensor toys confirm that anisotropic residuals preferentially activate trace-free tensor response.

• V11.1D confirms that m=2 quadrupole power tracks shear geometry and rotates with source orientation.

• V11.1E maps the V11.1B variables onto V11.7 scalar/tensor coefficients.

• The tensor channel high-k preference follows from the higher derivative order of div Q.

16. What This Note Does Not Establish

• The microscopic first-principles values of τ_χ, D_χ, α_χ, ε_χ, τ_Q, D_Q, α_Q, or ε_Q.

• A fully covariant relativistic stress tensor.

• A proof of Navier-Stokes regularity.

• A unique form for every possible lag-stress correction.

• A complete replacement for standard fluid dynamics.

The correct claim is narrower: V11.7 coefficients now have a defined SFT parent architecture, but their exact microscopic values remain open.

17. Earned Statement

The following statement is now structurally and computationally supported:

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The V11.7 fluid correction can be interpreted as the continuum limit of
delayed Planck-area ledger settlement.
V11.1 supplies the discrete write demand.
V11.1B supplies the finite-settlement residual.
V11.1D shows that anisotropic residuals preferentially load
tensor/quadrupole structure.
V11.1E maps the resulting coefficients onto the V11.7 scalar/tensor fluid
sector.
```

In compact form:

```
Fluid lag stress is unsettled boundary accounting made continuous.
```

18. One-Line Summary

V11.1 says one tick demands one Planck-area ledger entry. V11.1B adds that geometry may settle that demand with finite response time; the residual becomes scalar/tensor lag stress. V11.1D confirms the tensor/quadrupole structure, and V11.1E maps that structure directly onto the V11.7 fluid coefficients.